

What a necessity argument for $\hbar > 0$ must supply: a unit and an indeterminacy

Result, 2026-09-27. A positive universal floor on recorded comparisons in Newton’s Galileo comparison needs two separable ingredients. **(U)** a universal action unit, and **(I)** an indeterminacy of the record’s back-action, since the floors of the Planck paper (Theorems 5–6) bound *spreads* of disturbances, and a definite, later recoverable kick has zero spread. The two have opposite status.

- **(U) follows from classical radiation thermodynamics** (Theorem U). Kirchhoff’s universality, Wien’s scaling law, finite equilibrium energy density and low-frequency equipartition force the equilibrium spectrum to contain a universal constant β with units of kelvin-seconds. With Boltzmann’s constant it gives an action $\hbar = k_B\beta$, independent of every material constant. Its value is already in Wien’s displacement constant b : $\hbar = y_*k_Bb/c$, with y_* a pure number fixed by the spectral shape. No premise about the body, its mass or a commutator enters.
- **(I) is impossible classically** (Theorem I). Any model with Liouville dynamics of body and pointers, product preparations obeying a prior density ceiling, and Bayesian conditioning on pointer readings admits records whose body posterior has arbitrarily small area. A premise that yields (I) must deny one of these three. Quantum kinematics denies the third; a universal unrecordable background (stochastic electrodynamics) denies the second; an epistemic restriction in Spekkens’s sense imposes the ceiling directly. Newton’s printed optics denies none, which is paper §8’s $\neg M3$ as an instance.

So the necessity question becomes sharper than before. The existence of a universal action unit is forced by nineteenth-century physics that never mentions the body. What remains open is which of three explicit classical structures a physically justified premise must give up for that unit to act as a floor on records.

The candidate list and both theorems were proposed by a Fable reviewer at the user’s request (report in the session scratchpad, 2026-09-27), after three external models (DeepSeek, Mimosa, Muse Spark 1.1) converged on the Stoney unit k_e/c , which the dimensional note already holds. The ingredients are classical and the theorems elementary; no novelty is claimed for either. Their use here is to replace the obstruction map of action-scale-obstructions by one statement of what is missing.

1. Two ingredients

The floor of the Planck paper, Theorem 6, is

$$\frac{s}{8} \sum_j \Delta(\hat{D}_j) + \frac{J}{2} \sum_j \Delta(\hat{X}_j) \geq \hbar \arcsin(1 - 2\epsilon),$$

with Δ a spread of an undetermined disturbance. Its proof uses Mandelstam–Tamm in the Bures angle and the Weyl relations, that is, $\hbar > 0$ in two roles.

- **(U), the unit.** By Theorem A of the dimensional note, a floor shared by all masses and preparations is a pure number times a product of the model’s fixed constants with action units. Classical mechanics, gravity and electrodynamics hold exactly one mass-independent such product, $k_e/c = e^2/(4\pi\epsilon_0 c) = \alpha\hbar$.
- **(I), the indeterminacy.** The floor bounds spreads. A record whose kick is definite and later recoverable contributes nothing, whatever the size of the kick. For example, a charged probe passing a charged body at speed $u < c$ and impact parameter b delivers the transverse impulse $2n_1n_2k_e/(bu)$, so charge atomicity gives $|\Delta p|b \geq 2k_e/c = 2\alpha\hbar$:

a universal, mass-independent floor on the *magnitude* of a record's kick. The kick is determined by the reading, so its spread is zero. The right unit gives the wrong kind of quantity.

2. Theorem U: the unit from radiation thermodynamics

Let $u(\nu, T) d\nu$ be the equilibrium energy density of electromagnetic radiation between frequencies ν and $\nu + d\nu$ at absolute temperature T . Assume:

- **(K)** u is the same for every material enclosure (Kirchhoff 1859, from the second law: two cavities at one temperature with different spectra would drive an engine through a colour filter);
- **(W)** $u(\nu, T) = \nu^3 f(\nu/T)$ (Wien 1893, from adiabatic compression of a reflecting cavity and the second law (Wien 1893, reprint, metadata));
- **(S)** $\int_0^\infty u d\nu < \infty$ for every $T > 0$ (Stefan's measured law; Boltzmann's T^4 theorem from Maxwell's radiation pressure);
- **(R)** $u(\nu, T) \rightarrow (8\pi\nu^2/c^3) k_B T$ as $\nu/T \rightarrow 0$ (equipartition of the low-frequency modes, the regime Rayleigh and Jeans describe).

Theorem U. Under (K), (W), (S), (R) there is a constant $\beta > 0$ with units of kelvin-seconds, independent of every material constant, such that $f(\nu/T) = (8\pi k_B/c^3)(T/\nu) \varphi(\beta\nu/T)$ with φ a function of one dimensionless variable, $\varphi(0) = 1$. Hence $h := k_B\beta$ is a universal action, and

$$\sigma' = \frac{8\pi k_B^4}{h^3 c^3} \int_0^\infty y^2 \varphi(y) dy, \quad h = \frac{y_* k_B b}{c},$$

where σ' is the Stefan constant of the energy density, b is Wien's displacement constant ($\lambda_{\max} T = b$ for the peak of the spectrum per unit wavelength), and y_* is the pure number at which that peak sits.

Proof. By (W), u depends on ν and T through ν^3 and the ratio ν/T , whose units are $1/(\text{s} \cdot \text{K})$. By (K) the only constants u may contain are universal ones, and (R) displays two, c and k_B . From c , k_B and ν/T no dimensionless combination exists, because c contains a length and k_B an energy, and neither appears in ν/T . If u contained no further constant, f would be a pure power, and (R) would fix it to $f = (8\pi k_B/c^3)(T/\nu)$, whose frequency integral diverges, contradicting (S). So u contains a further universal constant. By (W) it can enter only through a dimensionless multiple of ν/T , so it has units of kelvin-seconds; call it β . Then f has the displayed form, (R) gives $\varphi(0) = 1$, and substituting $y = \beta\nu/T$ in the integral gives σ' . The peak of $u_\lambda = u c/\lambda^2$ lies at a fixed value y_* of $\beta c/(\lambda T)$, so $\lambda_{\max} T = \beta c/y_*$, which is the expression for $h = k_B\beta$. $\square$

With the spectral shape of Planck 1900 (Ann. Phys. 306, 69, metadata), $\varphi(y) = y/(e^y - 1)$, the integral is $\pi^4/15$, $y_* \approx 4.965$, and h is Planck's constant. The theorem needs no shape: any finite spectrum obeying (K), (W), (R) contains a universal action, fixed up to a pure number by the measured b , c and k_B . Planck formed his natural units from exactly this constant in 1899–1900, before the quantum hypothesis; the theorem states why a constant had to be there.

Two remarks keep the statement honest. First, k_B converts kelvins to energy, and its value needs atomism (the gas constant divided by Avogadro's number); the universal constant of (K)–(W) alone is β . Second, with h and k_e/c both available, Theorem A of the dimensional note no longer fixes a unique unit: any universal floor is $h G(\alpha)$ with $\alpha = k_e/(hc/2\pi)$ dimensionless. Uniqueness of the unit requires that the floor not depend on the charge, which holds for records made with neutral probes.

Bridge to the Opticks. Newton's *Opticks* holds, per colour, an interval of fits Λ and a corpuscle whose Λp is invariant under refraction (paper §§7–8), but no universality across colours:

on his ordering of corpuscle sizes, Δp decreases toward the violet with no least colour. Theorem U supplies exactly the missing cross-colour constant. Reading h as a per-corpuscle impulse, $p = h/\lambda$, needs more: Einstein’s fluctuation argument applied to an exponential spectral tail, which is the quantization of the field. So the unit enters honestly through (K)–(R), and quanta enter through the tail.

3. Theorem I: no classical ceiling on posteriors

Theorem I. Consider a classical model with (i) Liouville (Hamiltonian) dynamics of the body and its pointers, (ii) product preparations, each factor with a prior phase-space density at most $1/X$, and (iii) records formed by Bayesian conditioning on a pointer coordinate read to finite precision. Then for every $X > 0$ and $\eta > 0$ there is a protocol whose body posterior satisfies $\Delta q \Delta p < \eta$, with Δ the width of the support.

Proof. Let the body prior be uniform on $[0, a] \times [0, X/a]$ and the first pointer (y, p_y) uniform on $[0, \epsilon] \times [0, X/\epsilon]$; both have density $1/X$. The impulsive coupling with Hamiltonian $g q p_y$ acting for unit time maps $y \mapsto y + gq$ and $p \mapsto p - g p_y$, leaving q and p_y unchanged, and preserves the joint density by Liouville. Reading y to precision ϵ' confines q to width $(\epsilon + \epsilon')/g$. A second pointer (z, p_z) , uniform on $[0, \epsilon_2] \times [0, X/\epsilon_2]$ and coupled by $g_2 p_y p_z$, maps $z \mapsto z + g_2 p_y$ and kicks only y , which is already read. Reading z to precision ϵ'_2 confines p_y to width $(\epsilon_2 + \epsilon'_2)/g_2$. The body’s momentum is $p - g p_y$, so its posterior width is at most $X/a + g(\epsilon_2 + \epsilon'_2)/g_2$, and

$$\Delta q \Delta p \leq \frac{X(\epsilon + \epsilon')}{ag} + \frac{(\epsilon + \epsilon')(\epsilon_2 + \epsilon'_2)}{g_2},$$

which is below η for g_2 large and a large, or ϵ, ϵ' small. The prior ceiling only widens the unread conjugates p_y and p_z . $\square$

The second pointer reads the first pointer’s kick; this is paper §8’s countermodel (two position records recover the momentum) in mechanical form, and each classical entry of the obstruction map is a corner of it. The recoverable-delayed-record results C068–C069 are the corner $a \rightarrow \infty$, $\epsilon' \rightarrow 0$.

What (I) must deny. A premise that yields a universal posterior ceiling, and with it paper Theorem 6 with the ceiling in place of $\hbar$, must deny (i), (ii) or (iii).

- *Quantum kinematics* denies (iii): there is no joint phase-space density on which to condition, and every read is also a kick.
- *A universal background that no record can fully capture* denies (ii): every preparation is correlated with it. Stochastic electrodynamics is of this kind; Lorentz invariance fixes the spectrum of a random classical background as $\propto \nu^3$ up to one constant with action units (Marshall 1963; Boyer 1975; metadata). The ceiling then needs the extra clause that records cannot read the background itself, since otherwise Theorem I applies to the enlarged system.
- *An epistemic restriction* imposes the ceiling on every state of knowledge. Bartlett, Rudolph and Spekkens show that Liouville mechanics with such a restriction reproduces Gaussian quantum mechanics (PRA 86, 012103, 2012, metadata; Spekkens 2007 for the discrete toy theory). This is the posterior ceiling as a principle, with its value free; Theorem U then fixes the value up to a number.
- *Newton’s printed optics* denies none: its fits are determinate and its records conditionable, which is why paper §8 finds that what Newton printed entails $\neg M3$.

4. Consequence for STATE

STATE item 2 (Newton necessity) gains a sharp statement. The unit is settled classically by Theorem U. The open task is a physically justified premise that denies one of (i)–(iii) of

Theorem I, of which the three known routes are quantum kinematics, an unrecordable universal background and an epistemic restriction. The historical test worth doing next is the reviewer's second candidate: whether Newton's inflexion experiments (*Opticks* Book III, Obs. 1–11) give a bending law $\theta b \approx C\Lambda$ per colour. If they do, Newton's own optics denies (iii) for corpuscle records, per colour, and paper §8 must be reworded to say that $\neg M3$ follows only for inflexion-free records.